PLUG PULLER · TWO-SIDED PULLER

Quick start, two-sided puller

The shortest path from a stuck plug to a printed two-sided puller that fits it and your hand: get the file, measure, fill in the four Customizer steps, print, assemble, use.

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Which tool this is

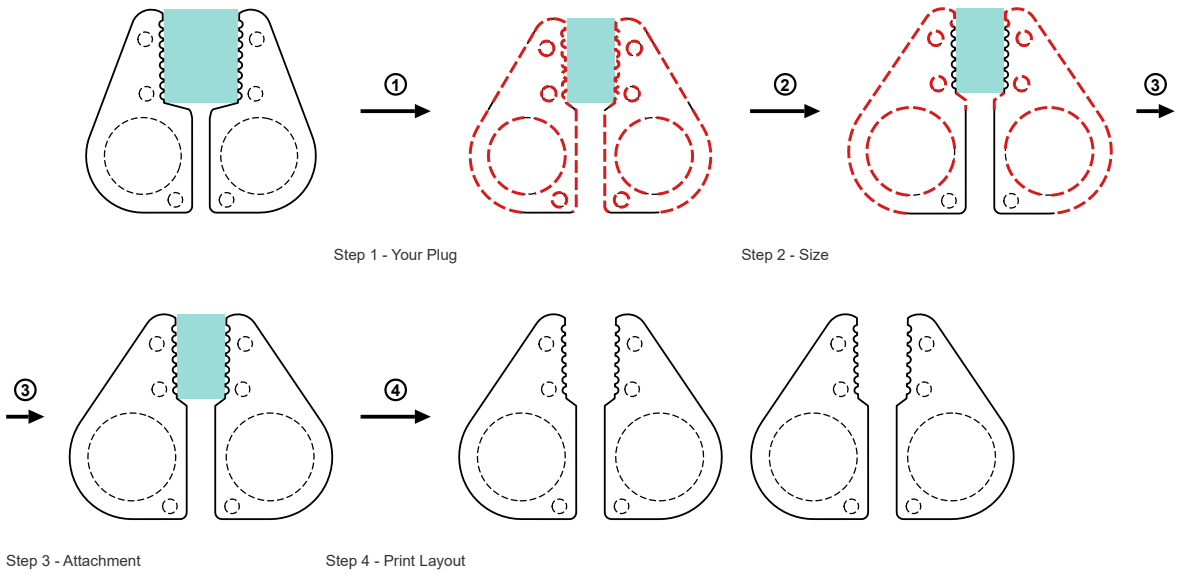
This guide is for the two-sided puller, src/Plug_Puller_Two_Sided.scad: two serrated plates that zip-tie around the plug and close across it, for a plug 24 mm thick or more and for a plug held from both sides, a USB-C tip or a round extension-cord plug. A thinner wall plug is the one-sided puller's job, and that tool has its own guide.

The Customizer's four steps are your plug, your size, the attachment and the print layout. This quick start covers getting the file, measuring your plug and your hand or matching a card, each step's dials in a table, then printing, assembly and use.

> If red text appears beside the part in the preview, read it: it names the measurement to fix, and the part will not fit until it is gone.

How to read the pictures

- **Black:** the tool
- **Red dashed:** the edges this dial moved
- **Teal:** your plug
- ① the numbers match the key beside the picture



Five stages of the two-sided puller, left to right, the top row first, the plug end at the top; arrows numbered 1 to 4 show the steps, and red dashes mark the edges each step moved. Start, the defaults: the tool as the file opens, with a 20 mm wide plug in teal. Step 1 - Your Plug: plug length 23 mm, plug width at both ends 13 mm, cord thickness 7 mm, plug sides Rounded sides; red on the arms, the finger lobes and the zip stations. Step 2 - Size: hand size Large; red on the arms, the finger lobes and the zip stations. Step 3 - Attachment: attachment Zip ties; no strap slot on a plug this short, so nothing moved. Step 4 - Print Layout: both plates side by side in one file, what you print; nothing marked.

- Step 1: type your plug's numbers and pick Rounded sides or Flat sides
- Step 2: pick your hand size
- Step 3: pick how it attaches; a plug this short gets no strap slot
- Step 4: both plates in one file

Get the file

The two-sided puller is the file `src/Plug_Puller_Two_Sided.scad`. There are two ways to open it and fill in its form: in your web browser with nothing to install, or in the free OpenSCAD program on your computer. Both show the same form, called the Customizer.

In your browser

The OpenSCAD Assistive Forge is a version of OpenSCAD that runs in your browser, on a computer or a phone, built for keyboard and screen reader use. Nothing is uploaded: your numbers stay on your device.

1. Open the two-sided puller in the Assistive Forge. On a first visit Forge asks which interface you want: choose **Assistive Forge**, then **Download & Continue**, and it downloads its engine once.
2. Forge offers to keep a copy of the project in your browser. Choose **Save My Copy**: the tool is then listed on Forge's main page whenever you come back, no link needed.
3. The form shows the four steps' dials first. The rest sit behind one button, **Show all parameters**.

If the link cannot load, Forge says why and offers **Try again**. You can also download the whole tool as one file, `dist/Plug_Puller_Two_Sided_SingleFile.scad`, and open it from Forge's main page, under Open or start a project.

On your computer

OpenSCAD is the free program that turns your numbers into a printable file. You use one panel of it and never touch the code.

1. Download OpenSCAD from https://openscad.org/downloads.html: the **Development Snapshot** for your system. This project is tested with the snapshot of 2026-01-03; any recent snapshot works. The regular release works too, only slower.
2. Get the project: on its GitHub page press the green **Code** button, then **Download ZIP**, and unzip it anywhere. Or download only `dist/Plug_Puller_Two_Sided_SingleFile.scad`, the whole tool in one file.
3. Open `src/Plug_Puller_Two_Sided.scad` (or the single file) in OpenSCAD: double-click it, or use **File ▸ Open**. A wall of code appears in an editor pane. Ignore it; you will not touch it.
4. Show the Customizer, the form you type into: in the **View** menu, make sure **Hide Customizer** is unchecked. In older versions it is **Window ▸ Customizer**.

The form's sections read top to bottom in the order you decide things: **Step 1 - Your Plug**, **Step 2 - Size**, **Step 3 - Attachment**, then **Step 4 - Print Layout**. Everything below Step 4 is optional.

Measure your plug and your hand

Everything here is in mm: a US plug is about 25 mm wide, so a 1 on your paper means you measured in inches. You need a caliper or a ruler with mm marks, the plug in its outlet, and your own hand only if you pick Measure my hand. Print the measuring form at 100 % and fill it in as you go: the sections below are the form's rows, in the same order, and the card names (R1, C1, F1 / F2) are the cards of the printed measuring stencil, described under Match a card instead of measuring.

The two-sided puller's widths are the size the two plates close across, so measure across the plug the way the plates will grip it. The thickness pair, the wall plate style and the hand width belong to the one-sided puller only.

Form row	What to measure	How	Typical	Stencil card
2	**Plug length** measure_plug_length	With the plug in the outlet, ruler from the wall plate face to the plug's back face. On a laptop or charger plug, measure from the device's edge.	15 to 55 mm; example 32.5 mm	card R1
3	**Plug width at the prong end** measure_plug_width_prong_end	Caliper across the plug body just behind the prongs, the size the two plates close across. On a USB-C or charger tip, just behind the metal tip; on a round plug it is the diameter.	10 to 40 mm; example 20 mm	card R1
4	**Plug head length** measure_plug_head_length	Ruler along the plug from the prong end to where the plug starts to narrow. If it narrows right from the prong end, type 0.	0 to 25 mm; example 15.2 mm	card R1
5	**Plug width at the cord end** measure_plug_width_cord_end	The same direction across the plates, measured at the plug's back end where the cord comes out. A rubber strain relief counts as part of the plug.	10 to 40 mm; example 11.1 mm	card R1
6	**Cord thickness** measure_cord_thickness	Caliper across the cord just behind the plug, the way the widths run: a flat cord's wide side, a round cord's diameter. On the stencil, it is the smallest C1 slot that takes the cord's wide side.	3 to 10 mm; example 7 mm	card C1
7	**Plug sides** plug_sides	Look at the plug body where the plates will grip it. Rounded sides for a round cord plug, a USB-C or charger tip; Flat sides for a boxy plug.	Choices: Rounded sides, Flat sides	none
10	**Finger width** measure_finger_width	Caliper across the widest knuckle of your middle finger, the finger that goes into the pull hole. With the stencil, find the smallest F1 or F2 hole your finger passes through comfortably and subtract 5 from its number. No caliper? Take a ring that fits that finger snugly, measure the ring's inner diameter in mm and add 1.5 mm.	16 to 24 mm; example 22 mm	card F1 / F2
12	**Strap width** strap_width	Read the width on the strap's packaging. ONE-WRAP comes in 10, 13, 16, 20 and 25 mm.	10 to 25 mm; example 15 mm	none

> Sanity check: each plug width is a two-digit number, roughly 12 to 45, and the prong-end width is usually the bigger one; the finger width is roughly 14 to 32. A number like 1.3 is inches: measure again with the mm side.

When you measure for someone else, a relative or a client, measure their hand for the finger and hand rows and their outlet and plug for the plug rows. Doubtful between two values? Round up: a slightly roomy fit works, a tight one does not.

The measuring form

> The form is printed at true size on page 00, at the end of this guide; the paper stencil sheets follow it. Print the form at 100 % (actual size, never fit to page) and check its 50 mm bar with a ruler before you trust it. Fill in the blanks top to bottom as you measure, then type the numbers into the Customizer in the same order.

Match a card instead of measuring

The measuring stencil is a set of thin printed cards. Each has a two-letter name raised on its face, so you can find it by touch. Hold your plug in the plug cards: if it fills one card's openings, snug with no big gaps, that card names the preset to pick in Step 1, and you type no numbers at all.

Card	What it is	What it answers
P1	The flat 2-prong lamp plug (NEMA 1-15)	the preset `Flat 2-prong lamp plug - NEMA 1-15`
P2	The standard 3-prong plug (NEMA 5-15)	the preset `Standard 3-prong plug - NEMA 5-15`
P3	The heavy-duty extension cord plug (NEMA 5-15)	the two-sided puller's preset `Heavy-duty extension cord - NEMA 5-15`
P4	The wide 2-prong appliance plug (NEMA 1-15)	the one-sided puller's preset `Wide 2-prong appliance plug - NEMA 1-15`
R1	A ruler: raised mm ticks, numbers every 10 mm, and a notch every 10 mm you can count by touch	the plug's length, widths and thicknesses
C1	A cord gauge: open slots 3 to 9 mm wide along one edge, slid onto the cord from the side	the cord thickness: the smallest slot that slips over the cord
F1, F2	Finger holes 15 to 25 mm across (F1) and 26 to 32 mm (F2)	your finger width: the smallest hole your middle finger passes through comfortably, minus 5

Each P card has three openings: W for the plug's width, T for its thickness, and an open slot that slides onto the cord. If no card fits, your plug is between presets: measure it instead. The measured tool fits better than any preset.

Printing the cards. Print `stl/Measuring-Stencil/Visual/Measuring-Stencil_Visual_All-Cards.stl`: 1.2 mm thick cards, no supports, any rigid filament, packed onto sheets for a 200 by 200 mm bed. The tactile version, `stl/Measuring-Stencil/Tactile/`, prints every label as a raised character at ADA size and adds a Grade 2 braille title flap to every card. The full guide covers a smaller bed and folding the flaps.

No 3D printer yet? The last pages of the printed quick start and full guide are paper versions of the cards at true size: the plug outlines, a 100 mm ruler and the finger circles. Print them at 100 % (actual size, never fit to page) and check the 50 by 50 mm square with a ruler before you trust them. On their own they are `stencil-sheet.svg` and `stencil-sheet-2.svg`.

The four Customizer steps

Work the form top to bottom. Each step lists its dials: what each one does, then its default and the choices or the range it allows. The full guide shows a picture of what every dial moves.

Step 1 - Your Plug

Pick a plug preset, or leave it on Measure my plug and type your plug's numbers; then pick Rounded sides or Flat sides.

Dial	What it does	Default and choices
Plug preset plug_preset	Fills in the plug's length, both widths, the head length, the cord and the sides from that plug's measurements, so the arms, the grip gap, the cord channel and the plate length all take that plug's shape at once.	**Default:** Measure my plug **Choices:** Measure my plug, Heavy-duty extension cord - NEMA 5-15, USB-C laptop tip, Flat 2-prong lamp plug - NEMA 1-15, Standard 3-prong plug - NEMA 5-15
Plug length measure_plug_length	Lengthens both arms so the teeth cover the whole plug body, and moves the zip-tie stations and the strap slot with them.	**Default:** 25.5 mm **Range:** 12 to 85 mm, in steps of 0.5 mm
Plug width at the prong end measure_plug_width_prong_end	Opens or closes the gap between the arms at their tips, where the plug's prong end sits.	**Default:** 20 mm **Range:** 5 to 40 mm, in steps of 0.5 mm
Plug head length measure_plug_head_length	Keeps the gap between the arms at its prong-end width for the length of the plug's head, so the gap narrows only behind the head.	**Default:** 0 mm **Range:** 0 to 60 mm, in steps of 0.5 mm
Plug width at the cord end measure_plug_width_cord_end	Opens or closes the gap between the arms at the plug's back end, so the arms taper to match the plug.	**Default:** 20 mm **Range:** 5 to 40 mm, in steps of 0.5 mm
Cord thickness measure_cord_thickness	Widens the cord channel between the finger holes, and the finger lobes move outward with it.	**Default:** 4 mm **Range:** 1.5 to 12 mm, in steps of 0.5 mm
Plug sides plug_sides	Removes the sloped cradle from the arms' gripping edges so the teeth bite straight along a flat-sided plug.	**Default:** Rounded sides **Choices:** Rounded sides, Flat sides
Show the see-through plug show_plug_preview	Shows or hides the see-through plug in the preview and changes nothing in the printed plates. Changes no shape: the see-through plug is a preview aid and is never exported.	**Default:** On **Choices:** On or off (a check box)

Step 2 - Size

Pick your hand size, or pick Measure my hand and type your finger width.

Dial	What it does	Default and choices
Hand size size	Widens both finger holes and their lobes, so the whole plate grows around them.	**Default:** Medium **Choices:** Small, Medium, Large, Measure my hand
Finger width measure_finger_width	Widens both finger holes and their lobes. Only acts when size is Measure my hand.	**Default:** 20 mm **Range:** 14 to 32 mm, in steps of 0.5 mm

Step 3 - Attachment

Pick how the tool attaches.

Dial	What it does	Default and choices
Attachment attachment	Removes the strap slot from each arm, while the zip-tie stations stay because they hold the plates together.	**Default:** Zip ties + Velcro strap **Choices:** Zip ties + Velcro strap, Zip ties
Strap width strap_width	Sets the strap the arm slot must clear, and when the slot's window between the zip-tie stations is shorter than the strap plus 1.5 mm the slot is left out.	**Default:** 15 mm **Range:** 10 to 25 mm, in steps of 1 mm

Step 4 - Print Layout

Pick whether the file holds both plates or one.

Dial	What it does	Default and choices
Print layout print_layout	Puts both identical plates side by side in one file, or just one plate.	**Default:** Both plates **Choices:** Both plates, One plate

Render, export and print

1. On the computer, press **F6** (or **Design ▸ Render**) and wait for the progress bar to finish; a render takes from a few seconds to a minute. In Forge, the preview draws on its own: wait for **Preview ready** under the picture.
2. Look for red text beside the part. If there is any, read it: it names the measurement to fix, and the part will not fit until it is gone.
3. Save the file: on the computer, **File ▸ Export ▸ Export as STL**; in Forge, **Export STL**. If you export without rendering, OpenSCAD asks you to render first: press F6 and export again. The quick preview (F5) looks complete but cannot be exported.

Load the STL into your slicer with these settings:

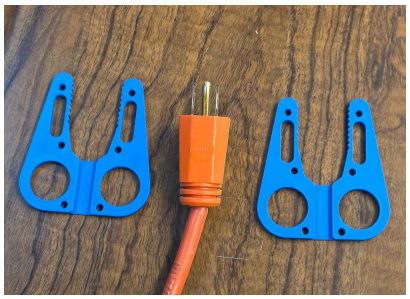
Setting	Value
Orientation	both plates flat on the bed, exactly as the file lays them out. No supports.
Layer height	0.2 mm
Walls	3 to 4: you pull hard on this part
Infill	25 to 35 percent
Material	PETG. PLA, ABS and ASA work too. Never a flexible filament: the tool has to stay stiff.

The file holds both plates side by side. Most slicers load it as one object with two parts, which prints fine as it is; use your slicer's split-to-objects command if you want to move one plate on its own.

Assemble

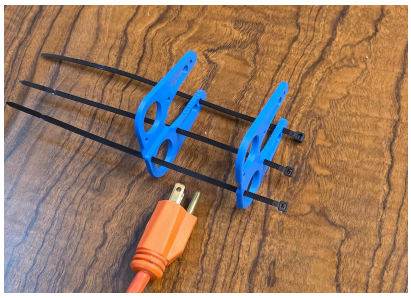
You need three zip ties; the size is in the bill of materials. The two plates are identical; one is flipped over so the toothed arms face each other.

1. Print both plates. The picture shows the two plates on a table with the plug between them.



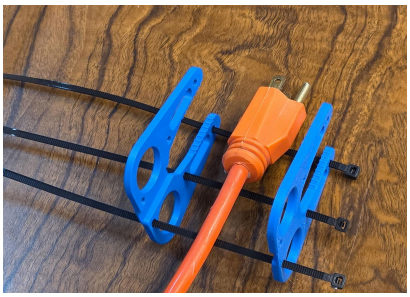
The two plates of the two-sided puller on a table with the orange extension-cord plug between them, ready to assemble.

2. Thread one end of each zip tie through the first plate, then through the second, so the ties lie between the plates. Make sure the cord channel of both plates faces the same way. The picture shows three ties through one plate, lying toward the other.



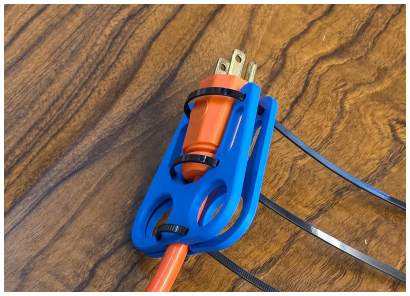
Three zip ties threaded through one plate of the two-sided puller and lying flat toward the second plate, the plug beside them.

3. Place your plug between the two plates, its prongs toward the arms. The picture shows the plug lying on the first plate between the ties.



The orange plug placed on the first plate between the zip ties, its prongs toward the plate's arms.

4. Thread the zip ties through the matching holes of the second plate. The picture shows the second plate over the plug with the ties through it and their tails loose.



The second plate laid over the plug, with the zip ties threaded through its matching holes and their tails hanging loose.

5. Pull the zip ties tight enough to hold the plug between the plates. The picture shows the ties cinched with their tails still on.



The zip ties pulled tight around both plates and the orange plug, their tails still sticking out.

6. Cut off the zip-tie tails. The picture shows the finished tool holding the plug.



The finished two-sided puller with the zip-tie tails cut off, holding the orange plug with its prongs clear.

The tool stays on the plug: the serrated arms do the gripping, and the zip ties hold the two plates together.

Use it

The plug puller is for anyone who cannot grip a plug and pull it out of an outlet: arthritis, low grip strength, tremor, a small hand, one hand. Two finger holes take the pull, so your whole hand does the work instead of a fingertip pinch. The tool touches only the plug's sides and back, never the outlet.

The two plates stay zip-tied around the plug, so there is nothing to seat: the tool is already on.

1. Put two fingers through the round holes at the cord end, one in each lobe.
2. Pull straight back, away from the wall.

The pictures show a two-sided puller and its plug seated in an outlet, ready to be pulled, and then the pull.



The two-sided puller and its orange extension-cord plug seated in a wall outlet, ready to be pulled, no hand on it yet.



A gloved hand pulling the two-sided puller and its orange extension-cord plug straight out of a wall outlet, two fingers through the holes.

A two-sided puller on a USB-C laptop tip works the same way, pulling the tip out of the laptop instead of a wall.

Safety

- Nothing goes between the plug face and the outlet cover: the tool grips only the plug's sides and back.
- Pull straight out. Never lever the tool sideways or up and down.
- Do not use the tool on a damaged cord, a cracked plug, or a plug that is warm to the touch.
- Keep your fingers away from the prongs as the plug comes out.
- The tool has no conductive parts, but it is still plastic near electricity: if it cracks, stop using it.

If it does not fit

Put the tool on the plug and pull once. The plug should sit flat between the plates with the cord in its channel, and your two fingers should slide in and out of the holes without catching. If anything is tight, loose or awkward, the full guide names the one dial to change and by how much, decodes every red warning tag, and covers the advanced dials: the two-sided puller's full guide.

Plug Puller — Measuring form, two-sided puller

Measure in mm. Fill in the blanks top to bottom, then type them into the Customizer in the same order.

Print at 100 % — 50 mm

1. `plug_preset`
   Plug preset
   - (●) **Measure my plug**
   - ( ) Heavy-duty extension cord - NEMA 5-15
   - ( ) USB-C laptop tip
   - ( ) Flat 2-prong lamp plug - NEMA 1-15
   - ( ) Standard 3-prong plug - NEMA 5-15

2. `measure_plug_length`
   Plug length
   default 25.5 mm → _____ mm card R1

3. `measure_plug_width_prong_end`
   Plug width at the prong end
   default 20 mm → _____ mm card R1

4. `measure_plug_head_length`
   Plug head length
   default 0 mm → _____ mm card R1

5. `measure_plug_width_cord_end`
   Plug width at the cord end
   default 20 mm → _____ mm card R1

6. `measure_cord_thickness`
   Cord thickness
   default 4 mm → _____ mm card C1

7. `plug_sides`
   Plug sides
   - (●) **Rounded sides**
   - ( ) Flat sides

8. `show_plug_preview`
   Show the see-through plug
   - (●) **leave on**

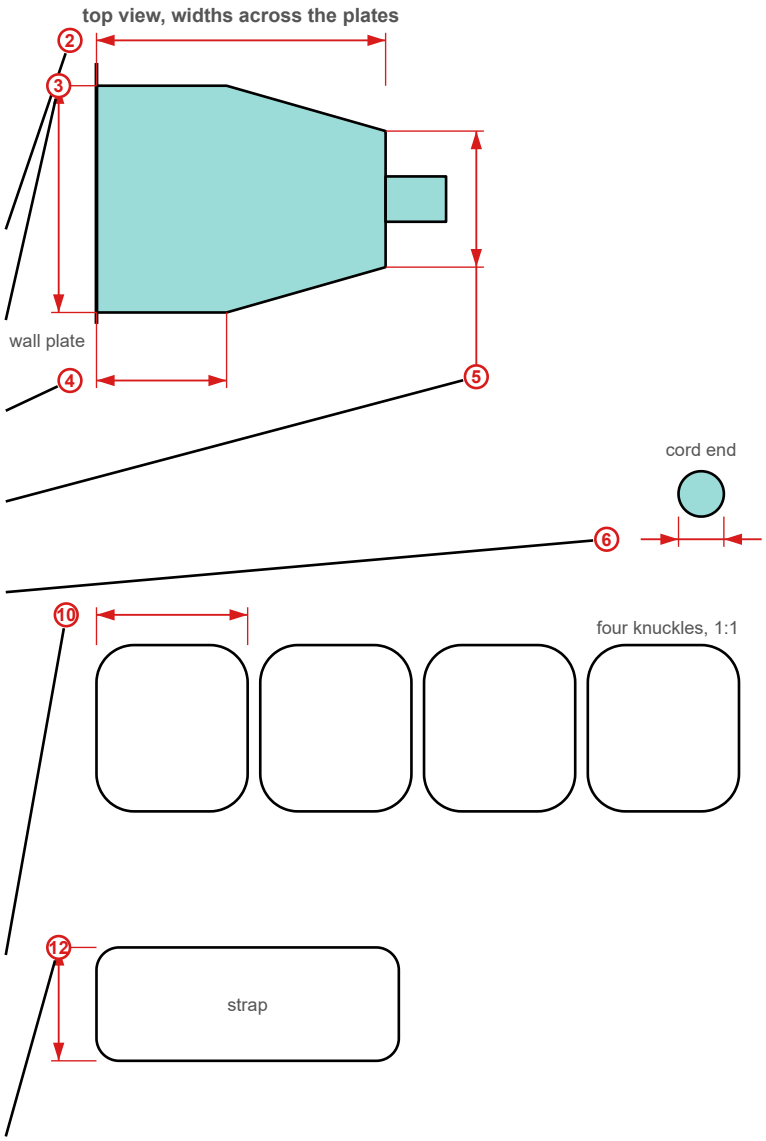
9. `size`
   Hand size
   - ( ) Small
   - (●) **Medium**
   - ( ) Large
   - ( ) Measure my hand

10. `measure_finger_width`
    Finger width
    default 20 mm → _____ mm card F1 / F2

11. `attachment`
    Attachment
    - (●) **Zip ties + Velcro strap**
    - ( ) Zip ties

12. `strap_width`
    Strap width
    default 15 mm → _____ mm

13. `print_layout`
    Print layout
    - (●) **Both plates**
    - ( ) One plate



Plug Puller — Measuring Stencil Sheet

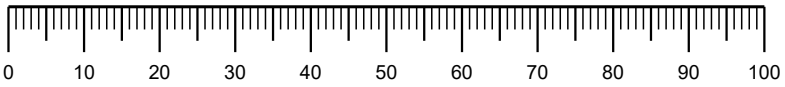
Print at 100% scale / “Actual size” — never “fit to page”. Check the calibration square before trusting anything on this sheet.

CALIBRATION
50 × 50 mm

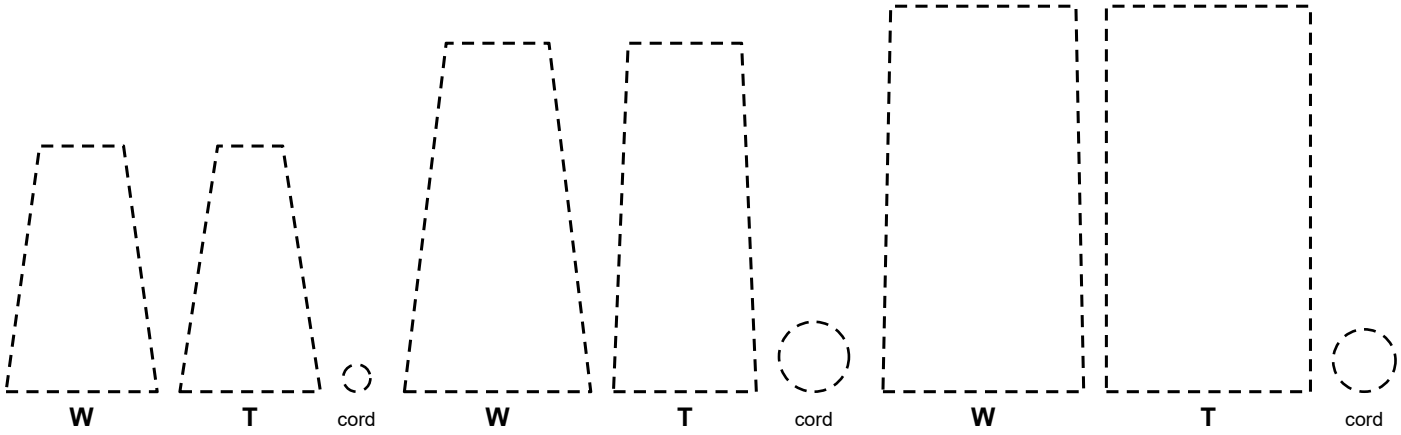
1. Measure this square with a ruler.
It must be exactly 50 × 50 mm — if not, your print was scaled: re-print at 100%.

Card legend: the quick start, under Match a card
Works on A4 and US Letter paper.

R1 — ruler (mm): plug length, widths, thicknesses, hand width



2. Plug silhouettes (1:1) — hold your plug against W (wide side) and T (thin side); if it matches, pick that preset in Step 1.



P1 Lamp 2-prong NEMA 1-15 **P2** Standard 3-prong NEMA 5-15 **P3** Heavy-duty cord NEMA 5-15

3. F1 / F2 — finger circles: smallest circle your middle finger passes through, minus 5 = finger width.

